4
Superpermeable Hydrogen Transport Membranes

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4.1
Introduction

Dense metal membranes fabricated from elements including niobium, tantalum and vanadium have been termed *superpermeable* by workers in the nuclear industry because the membranes, under certain conditions, are virtually transparent to hydrogen and its isotopes, having, in the limit, the same hydrogen flux as passes through an unobstructed aperture of the same cross sectional area [1–3]. Such dense superpermeable membranes are essentially 100% selective towards hydrogen, effectively blocking even the transport of helium. Superpermeable membranes have been used successfully to separate isotopes of hydrogen from helium in plasmas and to remove hydrogen isotopes from molten metal cooling fluids used in nuclear reactors.

Figure 4.1 plots the hydrogen permeability of selected elements, based upon the equations of Steward of Lawrence Livermore Laboratory [4]. The exceptionally high values for hydrogen permeability of niobium, tantalum and vanadium, which have long been of interest in the nuclear industry, have also attracted the attention of workers in the chemical industry, who have envisioned membranes based upon these elements for use in the extraction and purification of hydrogen from various chemical reactor streams. The potential for application of these elements in hydrogen transport membranes is reflected in a very extensive patent literature. Although the values for hydrogen permeability of Nb, Ta and V plotted in Fig. 4.1 are far superior even to those of palladium, these values fall far short from those predicted for true superpermeability. This chapter reviews the conditions under which true superpermeability has been achieved and the factors which prevent superpermeability from being realized in various applications.

Dense membranes for extraction and purification of hydrogen may be key components for the establishment of the hydrogen economy and for reduction of CO_2 emissions. Dense hydrogen transport membranes, which are essentially 100% selective towards hydrogen, may serve as a means for final clean-up of hydrogen before its use in proton exchange membrane fuel cells, which require hydrogen of the highest purity. Dense hydrogen transport membranes are envisioned for

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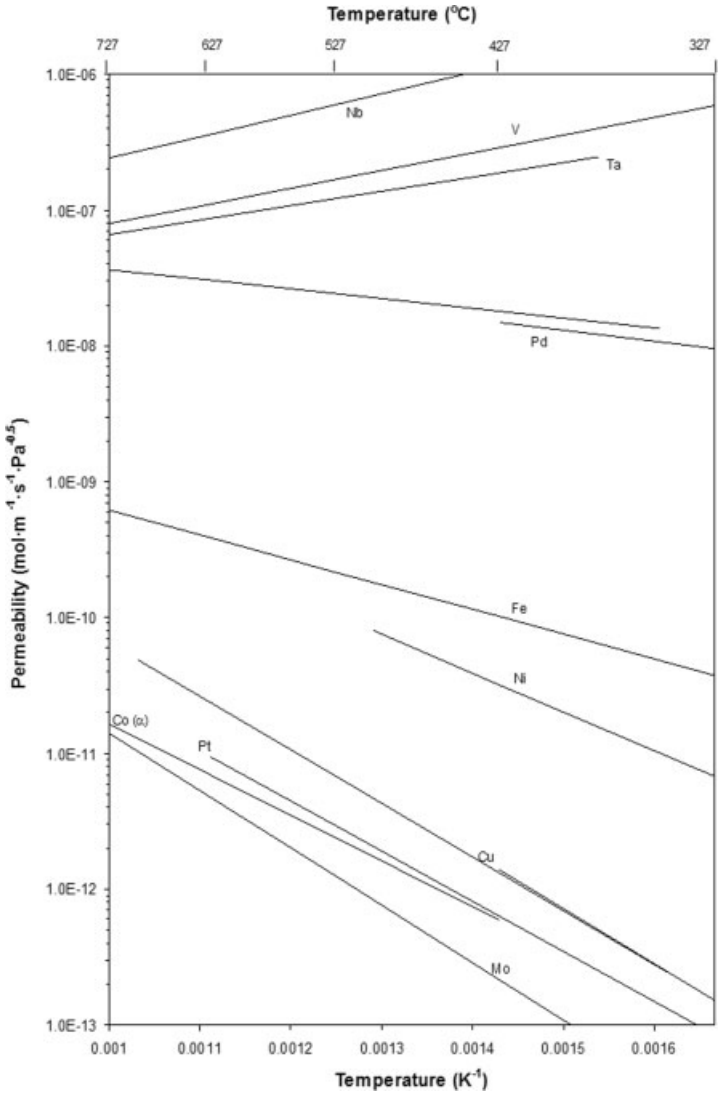


Figure 4.1 Plots of theoretical permeabilities for selected elements based upon the equations of Steward at Lawrence Livermore Laboratories [4].

extraction of hydrogen from reformates derived from methanol, natural gas and other hydrocarbon fuels, and for purification of hydrogen produced from the gasification of coal. Membranes have also been envisioned for use in the decomposition and extraction of hydrogen from H_2S. Hydrogen transport membranes may also serve to extract hydrogen from water-gas shift reactors for the production of clean hydrogen fuel while retaining CO_2 at high pressures (30–70 bar), which is

desired for economical sequestration of CO_2. The potential application of membranes for aiding sequestration of CO_2 is a major driving force behind the research and development of hydrogen transport membranes.

In this chapter, calculations based upon the kinetic theory of gases are first used to predict the hydrogen flux which, theoretically, might be achieved if superpermeability could be maintained over a wide range of membrane operating conditions. Next, conditions under which superpermeability has been verified in plasma physics and in the nuclear industry are reviewed. Factors which prevent superpermeability from being achieved in industrial chemical applications are discussed. Finally, the literature is reviewed to summarize the major issues involved in the practical application of these membrane materials and the progress which has been made. Major membrane research and development issues include protection of membrane hydrogen dissociation catalysts from catalyst poisons, minimization of interdiffusion between membrane hydrogen dissociation catalysts and membrane components, prevention of hydrogen embrittlement in both membrane and catalytic materials, and development of an economic method of scale-up and mass production. Progress has been made in the use of adsorbents to lower concentrations of catalyst poisons, the development of poison-resistant catalysts, and the development of new alloys, cermets (ceramic–metals) and other composite structures to resist hydrogen embrittlement.

4.2
Theoretical Limits of Superpermeable Membranes

For a superpermeable membrane, the flux of hydrogen through the membrane cannot exceed the number of gas phase hydrogen molecules incident upon the feed side surface of the membrane. This ultimate limit holds for all types of membrane, whether porous or dense. It is instructive, therefore, to calculate this limit for the hydrogen flux of superpermeable membranes, which, by definition, transport every atom of hydrogen incident upon the feed side surface of the membranes [1–3].

From the kinetic theory of gases [5–8], it is found that the number of molecules striking each unit area of surface depends upon the velocity of the molecules in the gas phase and the number of molecules per unit volume in the gas. For a simple estimate, the flux, J, incident upon a surface is given by:

$$J = \frac{1}{4} N\bar{c} = \frac{1}{4}\left(\frac{8RT}{\pi M}\right)^{\frac{1}{2}} = N\left(\frac{RT}{2\pi M}\right)^{\frac{1}{2}} = \frac{P}{RT}\left(\frac{RT}{2\pi M}\right)^{\frac{1}{2}} = P\left(\frac{1}{2\pi MRT}\right)^{\frac{1}{2}} \tag{4.1}$$

Where

J is the incident flux in mole m^{-2} s^{-1} or molecules m^{-2} s^{-1}
N is the density of the gas in molecules m^{-3}
$\bar{c}$ is the mean molecular velocity = $\left(\frac{8RT}{\pi M}\right)^{\frac{1}{2}}$ in m s^{-1}
R is the gas constant = 8.3145 J K^{-1} mol^{-1} = 8.3145 kg m^2 s^{-2} K^{-1} mol^{-1}
T is the absolute temperature in K

M is the molecular weight of the gas in kg mol^{-1} = 0.00201588 kg mol^{-1} for H_2
π = 3.14159…
P is the pressure in Pa = 1 kg m s^{-2} m^{-2}

As an example, a partial pressure of molecular hydrogen of only 1 torr (133.3 Pa) in the gas feed at 0 °C would produce a hydrogen flux of 24.9 (mol H_2) m^{-2} s^{-1} incident upon a surface or orifice. In other units, this flux is equivalent to 3.3 LH_2 min^{-1} cm^{-2} (STP), or to 1.5 × 10^{21} (molecules H_2) cm^{-2} s^{-1}. As a second example, a partial pressure of molecular hydrogen in the feed of 1 atm (101 325 Pa) at 0 °C would produce a hydrogen flux of approximately 18 900 (mol H_2) m^{-2} s^{-1}. In various other units, this is equivalent to 42.4 (L H_2) cm^{-2} s^{-1} (STP), 2540 (L H_2) cm^{-2} min^{-1} (STP), 3.81 (g H_2) cm^{-2} s^{-1} or 11.4 × 10^{23} (molecules H_2) cm^{-2} s^{-1}. The hydrogen flux would theoretically be greater still if the hydrogen partial pressures in the gas feed were increased above 1 atm.

These calculations, predicting extremely high hydrogen flux for superpermeable membranes, are strictly correct only for gas flow into a perfect hydrogen sink or into a perfect vacuum and assume that hydrogen molecules are not reflected backwards by intermolecular collisions [5]. They also assume electrically neutral molecules which are not accelerated by an electric field. In practice, true superpermeability has been achieved only under very special conditions, invariably with relatively low flux of hydrogen incident upon the membrane surfaces. Nevertheless, knowledge of the ultimate superpermeable flux leads to further inquiry concerning the identification of factors which inhibit membranes from achieving their theoretical flux and to the elimination or mitigation of the rate-limiting factors required for the practical improvement of membranes.

Under true superpermeable hydrogen flux conditions, the large numbers of molecules predicted to impact upon a membrane surface follow, in part, as a consequence of the very high molecular speeds of gas phase hydrogen relative to the size of reactor vessels. For example, the mean velocity of a hydrogen molecule, H_2, in the gas phase at 273 K (0 °C) is 1.7 km s^{-1} [8]. Mean molecular velocity increases in proportion to the square root of the absolute temperature. In a chemical reactor at 673 K (400 °C), for example, the mean velocity of H_2 will increase by a factor of (673 K/273 K)$^{1/2}$ from 1.7 km s^{-1} at 273 K to 2.7 km s^{-1} at 673 K. Mean molecular velocity decreases inversely with the square root of the molecular mass. For deuterium molecules, D_2, with a molecular mass approximately twice that of H_2, the mean molecular velocity is less than that of H_2 by a factor of $2^{1/2}$, approximately 1.2 km s^{-1} at 273 K (0 °C) [8].

As an interesting corollary of the inverse square root dependence of mean molecular velocity on molecular mass, it follows that if hydrogen travels through superpermeable membranes in a dissociated form (with a mass half that of molecular hydrogen), then the mean velocity of the dissociated hydrogen would be faster by a factor of $2^{1/2}$ relative to the mean velocity of molecular H_2 in the gas phase. This follows only if the membranes are truly superpermeable and if the dissociated hydrogen behaves as an ideal gas within the dense superpermeable membrane material. If the above is assumed true, it also follows that the flux of disso-

ciated hydrogen through superpermeable membranes would exceed the flux of arrival of molecular H_2 at the feed side surface of the membranes by a factor of $2^{1/2}$, and that the gas phase near the hydrogen feed side (retentate) surface would become depleted in molecular hydrogen. Near the hydrogen sink side surface of the membrane, the relatively slow moving hydrogen molecules in the gas phase would not move away from the surface of the membrane as rapidly as dissociated hydrogen is transported through the membrane, and hydrogen molecules would accumulate in the gas phase near the sink (permeate) side surface of superpermeable membranes. The depletion of gas phase hydrogen near the feed side surface and accumulation of gas phase hydrogen near the permeate side surface will reduce the concentration gradient of hydrogen across the membrane and, thus, the ultimate driving force for diffusion.

4.3 Superpermeable Membranes in Plasma Physics

Superpermeability has apparently been demonstrated for the extraction of hydrogen isotopes from plasmas when ion energies exceed 1 eV [1–3]. For example, Livshits et al., using a niobium tube, 1 cm in diameter, 18 cm long, and with thickness of 0.01 cm (100 microns), measured a plasma ion current density of 3 mA cm^{-2} incident upon the niobium membrane, which was electrically biased between –1 eV and –250 eV [2]. In this voltage range, the ion collection current was independent of the accelerating voltage, implying saturation even at 1 eV [2]. Using the measured ion current density of 3 mA cm^{-2}, and assuming, for simplicity, that the main plasma species is H_2^+, the calculated hydrogen flux incident upon the niobium membrane surface is: (3 mA cm^{-2})(1 A/1000 mA)(1 C s^{-1} / A)(1 mol/96485 C) = 3.1×10^{-8} (mol H_2^+) cm^{-2} s^{-1} (equivalent to 0.00031 (mol neutral H_2) m^{-2} s^{-1}). In various other units, this flux is also equivalent to 1.87×10^{16} neutral hydrogen molecules cm^{-2} s^{-1} or 0.042 (mL H_2) cm^{-2} min^{-1} (STP). This estimated hydrogen flux may vary if the concentrations of other species in the plasma, including H_2^+, H_3^+, D_2^+ and D_3^+, are taken into account. Nevertheless, the incident hydrogen flux from typical plasmas [9] will be relatively low and well within the capabilities of niobium, as well as many membrane materials not ordinarily classified as superpermeable.

Although the membranes used to extract hydrogen isotopes from plasmas meet the criterion for the definition of superpermeability, in that every atom of hydrogen incident upon the feed side surface of the membrane is transported through the dense membrane, it must be noted that superpermeability is achieved, in very large part, because of the relatively low flux of hydrogen incident upon the membranes. The plasma density in the quoted experiments was 5×10^{10} cm^{-3}, and the total gas pressure was relatively low, 0.002–0.004 torr (0.3–0.6 Pa) [2].

In the plasma experiments, the sticking coefficient of hydrogen on niobium must be equal to one if the membrane is to be classified as truly superpermeable. For highly energetic and ionized plasma species, including H_2^+, H_3^+, D_2^+ and D_3^+,

incident upon very reactive, clean, niobium surfaces, this appears to be a reasonable assumption. These reactive charged species apparently dissociate on niobium surfaces without the aid of catalysts. Presumably, reactive plasma species prevent the formation of thick layers of niobium oxides, carbides and nitrides, which otherwise might impede adsorption and diffusion of hydrogen and its isotopes. Surfaces of niobium membranes exposed to plasmas, however, are not entirely free of contaminants, as evidenced by oxygen and carbon detected by Auger electron spectroscopy [2].

It should be noted that the velocity of positively charged plasma species accelerated towards an electrically conductive membrane which is biased between –1 and –250 eV, will exceed that of the mean velocity of neutral molecules impinging upon a surface, as given by the kinetic theory of gases. For example, ions accelerated to only 1 eV possess a kinetic energy which is equivalent to the mean kinetic energy of gas molecules heated to 11 605 K (11 332 °C) [9]. Ions accelerated to only 1 eV will travel faster by a factor of $(11\,605\ \mathrm{K}/273\ \mathrm{K})^{1/2} = 6.5$, relative to neutral gas molecules at 273 K, which possess mean kinetic energy of approximately 1/40th of an electron volt [9]. Ions accelerated to 250 eV travel by a factor of $[250\ \mathrm{eV}/(1/40\ \mathrm{eV})]^{1/2}$ or 100 times faster than neutral gas molecules at 273 K. In the publications of Livshits et al., permeability through superpermeable membranes is stated to be independent of membrane temperature [2]. This appears to be possible considering that typical membrane operating temperatures are less than 800 K (527 °C), and that accelerated plasma ions have temperature equivalents in excess of 11 605 K (11 332 °C).

4.4 Hydrogen Transport Membranes in Nuclear Reactor Cooling Systems

Hydrogen transport membranes based upon niobium, tantalum, titanium, vanadium, zirconium and their alloys, have been used in the nuclear industry for the extraction of hydrogen and its isotopes from liquid alkali metal cooling fluids such as molten lithium, molten sodium, molten potassium, molten sodium–potassium eutectics, and other low-melting eutectics of the alkali elements. According to a patent of Hill [10], hydrogen dissociation catalysts on the membrane retentate surface are not necessary because molecular hydrogen is already dissociated in the molten metals to form alkali hydrides. Hydride (H^-) is transferred from the alkali hydrides to the retentate surfaces of membranes such as niobium or tantalum and their alloys [10]. A palladium catalyst, however, was used by Hill on the permeate side surface of the membranes [10]. In the work of Schulten et al., membranes fabricated from alloys of Zr, Ti, Ta and Nb are coated with iron on the molten metal side and with palladium on the permeate side [11]. The patent claims of Schulten et al. state that their membranes are capable of extracting isotopes of hydrogen from molten lithium, sodium, potassium and their eutectics held under pressures of up to 60 bar (870 psi) [11].

In the patent by Hill, an alloy of titanium containing 13 wt%vanadium, 11 wt% chromium and 3 wt% aluminum was developed as a hydrogen transport membrane material [12]. In this alloy, the crystal structure of titanium, which is normally hexagonal below 1153 K (880 °C), is stabilized in its high-temperature body centered cubic allotropic form. The body centered cubic crystal lattice is preferred for hydrogen transport. This titanium alloy was found to have hydrogen permeability superior to that of pure palladium in the range 300–450 °C (573–723 K) [12]. In addition, the titanium alloy was stated to possess a yield strength of 140 000 psi (0.97 GPa) at room temperature, ten times better than Pd, according to Hill [12].

In a patent of Behr et al., metallic Mg and Ca are added to molten alkali melts in order to getter oxygen and to protect membranes from oxide formation and oxygen embrittlement [13]. Niobium, tantalum, titanium, vanadium, and zirconium form extremely stable oxides. These very reactive metals can form oxides on their surfaces, which in thick layers inhibit the flux of hydrogen. The Group IVB and VB elements and some of their alloys are also susceptible to embrittlement by oxygen. Only a very few common oxides, including those of calcium, magnesium, aluminum, cerium, thorium, uranium and beryllium are more thermodynamically stable relative to the most stable oxides of Nb, Ta, Ti, V and Zr under typical membrane operating conditions. Of the common metals, only calcium, magnesium, aluminum and cerium are usually considered for reducing the most stable oxides of niobium, tantalum, titanium, vanadium and zirconium because of possible toxicity of oxides of uranium, thorium and beryllium.

For all hydrogen transport membranes, the partial pressure of hydrogen in the permeate must always be maintained at a partial pressure lower than that of the hydrogen in the source in order to maintain a net hydrogen flux from the source to the permeate [14]. For membrane applications with molten metal cooling fluids, in which upstream partial pressures in the retentate can be equivalent to only 2.7×10^{-2} Pa to 6.7×10^{-4} Pa (2.0×10^{-4} torr to 5.0×10^{-6} torr) [15], the partial pressure of hydrogen in the permeate must be kept exceptionally low.

One method of reducing the partial pressure of hydrogen in the permeate is to chemically react the hydrogen on the permeate membrane surface. According to the 1971 patent of Hill [10], hydrogen and its isotopes were catalytically reacted with oxygen to form water on the palladium catalysts coated onto the permeate side surface of the membranes. Thermodynamic equilibrium calculations, using $\Delta G = -RT\ln K_P$, where ΔG is the Gibbs' free energy of formation of water in the gas phase = –228.572 kJ mol^{-1}, and where K_P is the equilibrium constant, $(P_{H2O})/(P_{H2})(P_{O2})^{1/2}$, of the reaction $H_2(g) + {}^1/_2\, O_2(g) = H_2O(g)$, indicate that the partial pressures of hydrogen can theoretically be reduced to an immeasurably low value (~10^{-40} Pa), assuming 1 atm (101 325 Pa) partial pressure of oxygen in the permeate, and a vapor pressure of water of 23.8 torr (3170 Pa) after condensation at 298 K. The chemical method provides essentially the perfect hydrogen sink demanded by kinetic theory [5] for achieving maximum theoretical flux for superpermeable membranes. For comparison, turbomolecular pumps can, at best, reduce hydrogen partial pressures to only 1×10^{-8} to 1×10^{-9} Pa.

In another chemical method used to reduce the partial pressure of hydrogen and its isotopes in the permeate, Dittrich and Frey patent the use of getters of titanium, zirconium, hafnium or zirconium–aluminum alloys for the collection of hydrogen isotopes diffusing through membranes based on Nb, Ti, and Pd [16]. Hydrogen isotopes are released by heating the hydrogen getters. This method avoids the formation of water, but at a cost of a much lower driving force.

Because diffusion is simply rapid, random, translational motion of atoms and molecules, hydrogen in superpermeable membranes will travel just as quickly from the permeate side of the membranes to the retentate side, as it does in diffusing from the retentate to the permeate side. From the fundamental laws of diffusion, diffusion will tend to eliminate concentration gradients across a membrane, and at equilibrium the net flux across a membrane will be zero [14]. For superpermeable membranes, equilibrium will be very quickly approached, unless hydrogen is removed extremely rapidly from the permeate side of the membrane.

4.5 Hydrogen Transport Membranes in the Chemical Industry

In the chemical industry, it is desired to use hydrogen transport membranes for the extraction and purification of hydrogen from reaction mixtures containing partial pressures of hydrogen in the feed exceeding 30 bar (3 MPa). Therefore, the partial pressures of hydrogen in the permeate exiting chemical reactors may be kept much higher than in the examples from the nuclear industry while maintaining an economical flux (although not necessarily a superpermeable flux) of pure hydrogen. Partial pressure gradients across membranes can be maintained by chemical consumption of the hydrogen or by physical removal using a sweep gas or pumps and compressors.

In general, the ultimate driving force for diffusion of hydrogen across membranes is achieved in catalytic membrane reactors in which hydrogen is directly consumed in reactions on catalysts deposited onto the permeate side of the membranes [10, 17–19]. Reactions can include the catalytic hydrogenation of unsaturated organic compounds such as alkynes to alkenes, alkenes to alkanes, or aromatic compounds to cyclic alkene or alkane analogs. Other reactions include the reduction of nitrobenzene to aniline [18] or the partial reduction of organic acids, ketones, aldehydes and alcohols. Reduction need not be limited to organic reactions. For example, the formation of ammonia from molecular nitrogen might be envisioned. Hydrogen transport membranes provide a source of highly reactive dissociated hydrogen diffusing from beneath the catalyst surface. This may aid reactions normally self-poisoned by strong adsorption of reactants which otherwise would block coadsorption of molecular hydrogen.

If the catalytic conditions at the membrane permeate surface are not optimum for the desired chemical reaction, for example, due to constraints of membrane operating temperature dictated by the temperature of the retentate, desorbed molecular hydrogen may be consumed immediately downstream in catalytic reactors

in which the temperature or pressure conditions can be finely adjusted, heat exchange better controlled, or in which standard conventional high surface area catalysts can be employed. Chemical reactants for hydrogenation reactions may be used as sweep gases to dilute and reduce the partial pressure of hydrogen in the permeate and to carry the diffused hydrogen into reactors downstream.

Molecular hydrogen may be oxidized in fuel cells placed downstream from the membrane permeate surface, leading to hydrogen concentration gradients from the membrane permeate surface to the point of consumption. Steam, if available at low cost, or if desired downstream, may be used as a sweep gas. Pumps or compressors, designed especially for use with hydrogen, can also be used to create pressure gradients downstream from the membrane permeate surface.

It should be appreciated that for dense membranes it is only the partial pressure of hydrogen which needs to be lower in the permeate relative to the retentate to form a concentration gradient across the membrane. As pointed out by Pfefferle [20], whose patent discusses the use of nitrogen or steam as sweep gases, it is not necessary that the *total* pressure of mixed gases in the permeate be lower than that in the retentate. For dense membranes, the total pressure of mixed gases in the permeate could be equal to that of the retentate in order to eliminate mechanical forces across the membrane and to prevent rupture of thin membranes. If desired for a process application, the total pressure of mixed gases in the permeate could even be higher than that in the retentate, so long as the partial pressure of hydrogen in the permeate is kept lower than that in the retentate [20].

For superpermeable membranes, removing hydrogen from near the membrane permeate surface is a major challenge. For example, hydrogen gas would need to be removed from the permeate side of a superpermeable membrane at a rate of nearly 4000 (L H_2) min^{-1} cm^{-2} (STP) or 5.8 (g H_2) cm^{-2} s^{-1}, for a feed containing a partial pressure of hydrogen of only 1 atm (101 325 Pa) at a reactor temperature of 673 K (400 °C). For a partial pressure of hydrogen in the feed of 30 bar, the rate of removal would need to be approximately 30 times this or 120 000 (L H_2) min^{-1} cm^{-2} (STP).

Already by 1966, Straschil and Lopez realized that stagnant gas layers near membrane surfaces could limit the diffusion of hydrogen, even through thin palladium-based membranes [21]. This is especially true if the thin dense membranes are supported by porous layers with tortuous diffusion pathways. If hydrogen accumulates in the gas phase near the permeate side of the membrane, the concentration gradient across the membrane will be lowered and the driving force for diffusion reduced. To eliminate porous supports, they patented the use of punched-hole corrugated supports to allow maximum egress of hydrogen from the membrane permeate surface [21].

Other patented (nonporous) methods to provide support for thin membranes include the cylindrical screens for supporting thin membrane tubes of Bonnet [22], the metal gauze of Juda and Novack [23], the Dutch weave wire netting of Yamauchi, et al. [24], the circular screens of Bonnet [25], the perforated conduits of Eguchi et al. [26], the helical wires of Hollister and Stiles [27], the square perforations of Prölss [28], the corrugated screens of Iniotakis et al. [29], and the var-

ious supports of Edlund et al.[30, 31]. Although these methods lead to improved hydrogen flux relative to porous supports, they only partially ameliorate the problem of hydrogen accumulation in the gas phase near the permeate side of membranes.

Hydrogen diffusion through the gas phase can also be limited near the retentate side surface of the membranes. When dissociated hydrogen is transported more rapidly through dense superpermeable membranes relative to the rate at which gas phase molecular hydrogen reaches the feed side surface of the membrane, molecular hydrogen becomes depleted in the gas phase near the feed side surface of the membrane. This reduces the hydrogen concentration gradient across the membranes and thus the driving force for diffusion. Straschil and Lopez discuss the issues of poor gas mixing near the upstream surface of palladium-based membranes and problems with stagnant, or substantially stagnant, gas layers which inhibit contact of fresh feed with the upstream membrane surface [21]. Buxbaum was also aware of gas phase mass transfer issues in the gas feed, especially for membranes of niobium, tantalum and vanadium [32]. He patented methods to improve mixing and turbulence in the gas feed to minimize the thickness of the stagnant layer near membrane feed surfaces [32]. These included the use of pellets, packing, bends, catheters, baffles and protuberances to generate turbulence [32]. In the application of turbulence generators, especially with pellets and packing materials, care must be exercised to minimize pressure drops to the membrane, which would negate advantages of increased gas mixing.

4.6
Membrane Hydrogen Dissociation Catalysts and Protective Layers

For hydrogen transport membrane applications of interest to chemists and chemical engineers, hydrogen dissociation catalysts on both membrane surfaces (see Fig. 4.2) are of critical importance. Sieverts' Law, which shows a square root dependence of hydrogen flux on hydrogen partial pressure, implies that transport of hydrogen through dense metals, including palladium, niobium, tantalum, vanadium, zirconium and titanium, occurs in a dissociated form [33]. For membrane use in applications other than plasmas or molten alkali metal cooling fluids, in which hydrogen species are highly energetic or pre-dissociated, this implies that molecular hydrogen must be adsorbed and catalytically dissociated on the feed side surface of the membrane. The dissociated hydrogen, after diffusion through the dense membrane material, must be recombined and desorbed as molecular hydrogen from the permeate side of the membrane.

In principle, all atomically clean transition metals will adsorb and dissociate molecular hydrogen. For noble metals such as palladium, which do not readily form stable oxides, carbides and nitrides under membrane operating conditions, no additional catalyst or protective layer is normally required. However, for the less noble metals, and especially for highly reactive metals including Nb, Ta, Ti, V and Zr, surface layers of oxides, carbides and nitrides will rapidly form and poison

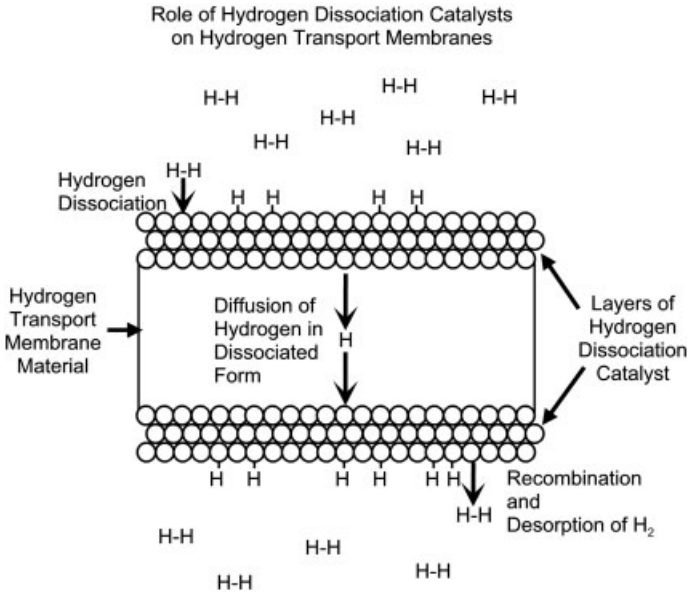


Figure 4.2 Schematic representing the need for catalysts on the retentate side of the membrane for adsorption and dissociation of molecular hydrogen and for catalysts on the permeate side of the membrane for recombination and desorption of molecular hydrogen into the permeate. Hydrogen is transported through the dense catalyst layer and through the bulk membrane material in a dissociated form.

the hydrogen dissociation reaction. Furthermore, these metals can react to form brittle bulk oxides, carbides and nitrides, which destroy the membranes. Such chemically reactive membrane materials must be coated with protective catalytic layers.

A number of issues must be considered regarding the use of protective catalyst layers on superpermeable membranes. The bulk hydrogen permeability of the dense, continuous, catalyst layers must be sufficiently high or the catalyst layers must be sufficiently thin, so as not to limit the hydrogen flux to levels which are not economically acceptable. The catalyst layers must adequately adhere to the substrate and must be in intimate contact with the substrate at the atomic level in order to allow transfer of dissociated hydrogen. The match in coefficients of thermal expansion and in chemical expansion between catalyst layers and membrane substrate must be sufficient to minimize interfacial stress, to prevent formation of major defects such as tensile cracks and to prevent delamination. The relatively small surface area of the membrane catalysts, relative to commercial supported catalysts, makes the membrane catalysts highly susceptible to catalyst poisons. Membrane catalysts must be designed for resistance to anticipated catalyst poisons, or catalyst poisons must be reduced to sufficiently low concentrations by use of adsorbents or other means to avoid poisoning of catalysts on the membrane surfaces. Substrate surfaces must be sufficiently clean prior to deposition of the

catalysts, and interfaces must be kept sufficiently clean during deposition of the catalyst layers. The interfaces between the catalyst layers and substrates must not be allowed to accumulate impurities which can lead to lack of wetting of the catalyst layers or to delamination of the catalyst layers. Impurities at the interfaces between the catalyst layers and the substrates must not act as hydrogen diffusion barriers, blocking permeation of hydrogen to levels less than those which are required for economic operation. Finally, issues of interdiffusion between components of catalyst and substrate layers must be addressed.

Common hydrogen dissociation catalysts, which can also serve as protective coatings, include the more noble metals such as palladium, platinum, iridium, and rhodium along with their alloys and intermetallic compounds, and to a lesser extent nickel, cobalt, copper and their alloys. These elements do not readily form carbides or nitrides and do not readily form stable oxides under the highly reducing atmospheres to which hydrogen transport membranes are exposed. For use as continuous protective films on Nb, Ta, Ti, V and Zr, palladium and its alloys have been most commonly employed because of the good combination of properties for hydrogen adsorption and dissociation, high hydrogen permeability, and chemical stability.

In 1964, Makrides et al. filed a patent (issued in 1967) on improved hydrogen permeable membranes fabricated by coating palladium and its alloys onto both sides of thin foils of vanadium, tantalum and niobium [34]. Stated advantages relative to pre-existing membranes of palladium and palladium–silver included lower cost and ability to withstand greater differential pressure due to the much greater tensile strength of the Group VB metals. Foil thickness of 10 mils (250 μm) was commonly employed in practice. According to Buxbaum and Kinney, membranes of the type patented by Makrides et al. were operated continuously for over 10 000 h (416 days) to remove hydrogen from nuclear reactor molten metal cooling fluids [35].

In one embodiment patented by Makrides et al., tantalum foils were electrolytically etched in hydrofluoric acid, washed with acetone and, while still wet with acetone, placed in vacuum and dried by evacuation. Using an argon plasma at a pressure of about 1 mm Hg (133.3 Pa), palladium was deposited onto both sides of the membranes to recommended thicknesses between 10 and 100 nm. Membranes of niobium and vanadium were prepared in the same manner, except that, in addition, vanadium was degassed in vacuum at 1273 K (1000 °C) to remove oxygen. Unalloyed palladium as well as Pd–Ag, Pd–Au and Pd–B, were also patented as hydrogen dissociation catalysts and as protective layers for the highly reactive niobium, tantalum and vanadium.

For the preferred composition of coatings of palladium–silver alloys, the patent of Hunter [36] is referenced by Makrides et al.. In the Hunter patent, maximum hydrogen permeability through membranes of Pd–Ag was obtained using a composition containing 27 wt% silver.

4.7 Thermal and Chemical Expansion

According to the patent of Makrides et al., the preferred operating temperature of the composite membranes was 673 K (400 °C). In order to avoid hydrogen embrittlement of both the palladium-based coatings and the Group VB elements during start-up and shut-down, hydrogen was purged from the systems before heating or cooling in the range from room temperature to 583 K (310 °C). This is in accord with previous work of Darling [37], who stated that palladium membranes must be maintained above 583 K (310 °C) in the presence of hydrogen in order to avoid formation of the brittle beta phase palladium hydride, which swells the palladium fcc cube-edge lattice parameter by 3.2% from 0.3894 to 0.4018 nm at room temperature. Already by 1952, Jost had reported that palladium samples, coated with palladium black to increase the surface area for catalysis, would swell and rupture at low temperatures when the concentration of hydrogen in the palladium lattice reached a value corresponding to $PdH_{0.54}$ [14].

It follows from the work of Darling and others that formation of the beta phase palladium hydride must be avoided to prevent embrittlement, cracking and delamination of thin catalytic films of unalloyed palladium used on composite membranes of Nb, Ta, Ti, V, and Zr. This also holds true for alloys of palladium susceptible to hydrogen embrittlement. Likewise, transformation of Nb, Ta, Ti, V and Zr substrates into brittle hydrides at low temperatures or at high hydrogen partial pressures must also be avoided.

Already by 1963, for a patent granted in 1966, Straschil and Lopez realized that the match of coefficient of thermal expansion between palladium membranes and (porous) substrates was critical, and stated that it would be virtually impossible to compensate for differences in dilation due to absorption of hydrogen [38]. They patented the use of dimpled or corrugated foils to accommodate differential thermal and chemical expansion [38]. Buxbaum and Hsu, in a 1992 patent, maintained that a rough substrate surface produced by abrasion with steel wool was critical for adherence of palladium on surfaces of Nb, Ta, V and Zr [39]. Other patents recommend corrugated or undulating configurations to allow for both thermal and chemical expansion [24, 26, 27, 29].

Table 4.1 lists coefficients of thermal expansion of unalloyed Pd, Nb, Ta, V and Zr, at various temperatures, compiled from data in the handbook edited by Touloukian et al. [40]. Table 4.1 also lists the percentage of thermal mismatch at various temperatures. The percentage of thermal mismatch was calculated by taking the values of the coefficient of thermal expansion of: [(Pd Overlayer –Substrate)/Substrate] ×100%. The palladium–vanadium system has the best overall match of thermal expansion in the membrane operating range of interest. Thermal expansion matches between palladium and tantalum and palladium and zirconium are poor, implying that, upon heating, palladium films will expand at a much greater rate than the substrates and contract at greater rates upon cooling. This could lead to film failure and delamination, as, for example, observed by Rothenberger et al. for 1.2 μm thick unalloyed palladium films sputter coated onto unalloyed tanta-

lum substrates, after 49 h use at 773 K (500 °C) [41]. As discussed, it should be appreciated that interfacial contaminants located between the substrate and the catalytic overlayer can also lead to delamination.

Table 4.1 does not take into account the effects of chemical expansion caused by absorption of hydrogen into both overlayer and substrate. Chemical expansion is a complex function of the solubility of hydrogen in the membrane materials. In the nonequilibrium steady-state operation of membranes, chemical expansion depends on temperature, the partial pressure of hydrogen on both sides of the membrane, the hydrogen feed rate and permeate removal rate, membrane and catalyst thickness, etc. In general, chemical expansion of unalloyed vanadium by dissolved hydrogen is least relative to unalloyed Ta, Nb and Zr. In principle, greater chemical expansion of the latter three elements could compensate for the lower thermal expansion relative to palladium and make the match better than the values listed in Table 4.1 might imply. In general, however, high chemical expansion is deleterious. Thinner films will have lower total thermal and chemical induced stress compared to thicker films. It should be noted that use of alloys in the palladium films with alloys of Nb, Ta, V and Zr in the substrates provides more flexibility for better matches in both thermal and chemical expansion or for minimization of chemical expansion relative to the unalloyed elements. In all cases, the solubility of hydrogen in both the catalytic layers and the substrates must not exceed values which transform the malleable metallic materials into brittle hydrides.

Tab. 4.1 Thermal linear expansion of palladium compared to Nb, Ta, V, Zr ($\alpha \times 10^6$) and thermal mismatch [(overlayer-substrate) / (substrate)] × 100%.

Temperature (K)	Pd	V	Pd–V%	Ta	Pd–Ta%	Nb	Pd–Nb%	Zr	Pd–Zr%
50	3.5	1.3	169	2.6	35	2.0	75	1.6	119
100	8.0	5.1	49	4.8	67	5.2	54	4.0	100
200	10.7	7.1	51	6.0	78	6.8	57	5.2	106
293	11.8	8.4	40	6.3	87	7.3	62	5.7	107
400	12.6	9.6	31	6.6	91	7.4	70	5.9	114
500	13.2	9.9	33	6.8	94	7.8	69	6.6	100
600	13.6	10.2	33	6.9	97	8.0	70	7.1	92
700	14.1	10.5	34	7.1	99	8.1	74	7.6	86
800	14.5	10.9	33	7.2	101	8.2	77	7.9	84
900	15.0	11.2	34	7.2	108	8.4	79	8.0	88
1000	15.6	11.6	34	7.3	114	8.6	81	8.2	90

4.8 Methods of Catalyst Application

The catalyst deposition method is critical to the success of the composite hydrogen separation membranes. Techniques for coating Nb, Ta, Ti, V and Zr with protective catalyst layers are not limited to the argon plasma sputtering method of Makrides et al. [34]. In the patent of Hill [10], a solution of palladium chloride dissolved in hydrochloric acid, to which ethanol and turpentine as reductant are added, is used to deposit palladium onto the inside of membrane tubes. In a later patent, Hill includes sputter deposition of 0.5 μm (500 nm) thick layers of Pd onto Ti alloys along with improved methods of electroless deposition. Hill stresses the importance of substrate surface preparation before the catalyst is applied [12]. In one method, Hill includes argon sputtering to remove oxide films or other surface hindrance to the entry or emergence of hydrogen [12].

Buxbaum et al. use an electroless method to deposit palladium onto disks and tubes of Nb, Ta, V and Zr [39, 42–45]. In one patent, Buxbaum recommends the use of thin palladium films coated with palladium or platinum black [44]. The thin palladium films protect the underlying substrates, while not greatly limiting bulk hydrogen diffusion, whereas the palladium or platinum black increases greatly the surface area of the catalysts, increasing hydrogen flux in cases where the flux is limited by adsorption or surface catalysis. *Platinum* black is recommended over palladium black for increased resistance to poisoning by sulfur. Substrate surface preparation is critical for the success of the catalyst deposition. Buxbaum et al. recommend abrasion with steel wool or a steel wire brush to remove thick adherent oxide layers before electroless deposition. Considering that the refractory oxides, carbides and nitrides of Nb, Ta, V and Zr are extremely difficult to remove by conventional chemical means, abrasion is a sound approach if thick contaminant layers on substrates are initially present before catalyst deposition.

It must be appreciated, that surfaces of Nb, Ta, V and Zr will be immediately re-oxidized or otherwise contaminated upon re-exposure to air, albeit with thinner layers (or in the limit a monolayer) relative to thick native films often encountered before abrasion. As a minimum, monolayer adsorption of contaminants is almost certainly assured in all but the best ultra-high vacuum ($< 1 \times 10^{-10}$ torr or $<1 \times 10^{-8}$ Pa) or in systems which simultaneously sputter substrate surfaces with argon or other inert ion during catalyst deposition. Some interfacial impurities between the substrate and the catalyst layer are tolerated in practice. However, the state of the substrate surface immediately prior to catalyst deposition is critical for wetting and adherence of the catalyst layers and for prevention of delamination. Theoretical flux maxima will not be achieved if thick impurity layers at the catalyst/substrate interface hinder hydrogen diffusion.

Edlund et al. have used thin foils of palladium or platinum laminated onto Nb, Ta or V substrates. [46–52]. Edlund and Pledger used platinum–vanadium–palladium laminated composites at 973 K (700 °C) to decompose H_2S at partial pressures up to 115 psia (7.9 bara) and total pressures up to 1000 psi (69 bar) [48]. Removal of hydrogen through the composite membranes aids the decomposition

of H_2S. The Pt and Pd foils were each 25 microns thick [46]. Platinum was used on the retentate side of the membrane exposed to H_2S and showed good resistance towards poisoning by sulfur at the temperatures employed, according to Edlund and Pledger [46]. Although sulfur is a notorious poison for platinum catalysts used at low temperatures, the sulfides leading to poisoning appear not to be stable at 973 K (700 °C) and the reaction conditions employed by Edlund and Pledger. Palladium was used on the permeate side but could not be used on the retentate side due to rapid catastrophic reaction with sulfur [46, 48]. At the relatively high temperatures used 973 K (700 °C) interdiffusion between palladium and vanadium was problematic, according to Edlund and McCarthy [47]. To minimize interdiffusion, diffusion barriers were introduced between the noble metal catalyst layers and the substrates. The diffusion barriers included proton-conducting perovskites, aluminum oxide felt, and a variety of oxides, sulfides and other materials [31, 46–52].

A group at Los Alamos has used e-beam (electron beam) evaporation of pure palladium onto tantalum, after cleaning the latter by argon ion bombardment [53]. They have also used ion sputtering. Both are performed in a vacuum chamber with base pressure of 1×10^{-6} torr (1.3×10^{-4} Pa) [54]. Copper gaskets and flanges of the type used in ultra-high vacuum technology are used in membrane permeation tests to ensure leak-free seals [53]. In their patent, Dye and Snow state that e-beam evaporation and ion beam sputtering are their preferred methods of catalyst deposition [55]. Substrates are ion milled with argon, neon or helium in a chamber evacuated to relatively modest base pressures between 10^{-3} torr and 10^{-6} torr (0.1 to 0.0001 Pa) [55]. The lower base pressure is preferred [55]. Membranes of V, Nb, Ta, Ti and Zr are coated with Pd, Pt, Ni, Rh, Ir, Co and alloys thereof [55]. Diffusion barriers of oxides, sulfides and selenides, for use at the catalyst/substrate interface, are patented [55]. Paglieri et al., sputter coat Pd, $Pd_{0.50}Ag_{0.50}$ or $Pd_{0.47}Cu_{0.53}$ at a rate of 0.3–0.5 nm s^{-1} to thicknesses of 30 to 250 nm onto foils of $V_{0.95}Ti_{0.05}$, $V_{0.88}Cu_{0.12}$, $V_{0.99}Cu_{0.01}$, and Ta in a vacuum system with base pressure of 1×10^{-7} torr (1.3×10^{-5} Pa) [56]. Nickel gaskets are used to form leak-tight seals in permeation tests [56].

Hara et al., use a radio-frequency (RF) sputter deposition technique to deposit 100 nm thick films of palladium onto amorphous alloys of $Zr_{36}Ni_{64}$, $Ti_{39}Ni_{61}$ and $Hf_{36}Ni_{64}$ [57, 58]. Eutectic alloy compositions are preferred for ease of melting and ease of forming amorphous alloys upon rapid quenching of molten alloys onto cold rotating plates.

The group of Nishimura at Tsukuba has employed a wide variety of deposition techniques in their membrane work [59–68]. In their early studies, palladium was electroplated onto both sides of the substrates [61]. Electroless deposition using $PdCl_2$ with $SnCl_2$ sensitizers was also used [59, 60]. The authors recommend abrasion using 0.05 μm (50 nm) diameter alumina particles to remove thick native layers of contaminants after alloy fabrication and rolling of foils [60]. It should be noted that workers skilled in the art avoid abrasives such as diamond paste or silicon carbide (SiC) to minimize possible formation of refractory carbides or silicides with the reactive substrates.

The group of Nishimura has evaporated 100 nm thick films of palladium onto both sides of foils of vanadium alloyed with nickel, aluminum and molybdenum in an ultra-high vacuum system equipped for physical vapor deposition [62–66]. After catalyst deposition, the membranes are placed in reactors, and both sides of the membranes are exposed to oxygen [60]. Reaction with oxygen presumably removes carbon and sulfur impurities from the palladium catalyst surfaces as well as from the reactor wall surfaces. If not removed these could poison the palladium catalysts and limit hydrogen flux. Palladium forms a coherent oxide, which expands only by about 1.6% relative to metallic palladium. This expansion is less than that of the beta-hydride, which expands by about 3.2%.

The group of Nishimura has studied alloys of vanadium nickel. Nickel lowers hydrogen solubility relative to that of unalloyed vanadium, and reduces the susceptibility of membranes to hydrogen embrittlement. This allows the use of membranes at much lower temperatures than is feasible with unalloyed vanadium. Aluminum in the vanadium alloys, present as a residual impurity from the thermite-type process used to reduce vanadium oxides, may be essential for gettering residual oxygen and reducing the susceptibility of vanadium to oxygen embrittlement. Molybdenum in the alloys may stabilize grain boundaries in vanadium and thus increase membrane mechanical strength.

Nishimura et al. have also co-deposited 100 nm thick films of palladium–silver alloys at a rate of 0.1 nm s^{-1} onto V–15 Ni and V–Ni–Al alloy substrates using separate Pd and Ag sputtering sources in a system with base pressure of 8×10^{-6} Pa (6×10^{-8} torr) [67, 68]. Using a coating of Pd–30 Ag (atom%), the hydrogen flux was increased 40 to 43% relative to composite membranes coated with unalloyed palladium [67, 68]. Moreover, an operating temperature as low as 423 K (150 °C) was achieved with V–15 Ni substrates without hydrogen induced embrittlement of overlayers or substrates. Lower operating temperatures and the use of alloys in both overlayer and substrate, should greatly reduce problems of interdiffusion, which was noted by other researchers working at higher temperatures [47, 56]. In the work of Nishimura et al., Pd–30Ag alloys produced slightly higher permeability in the composite vanadium-based membranes relative to the conventional Pd–25 Ag (atom%) composition [66]. In bulk Pd–Ag membranes used without the vanadium alloys [66], however, the Pd–25 Ag composition allowed the highest hydrogen flux.

As with all catalysts, palladium and its alloys are susceptible to poisoning [69]. Catalysts must be designed with resistance to poisoning, and proper precautions must be taken to minimize exposure of the membranes to catalyst poisons [69]. Typical poisons for palladium include H_2S and other compounds of sulfur such as carbon disulfide (CS_2), carbonyl sulfide (COS), aromatic thiophenes and mercaptans (thiols, R–SH). Palladium is poisoned by the Group VA elements, P, As, Sb and Bi, the halides (Cl, Br, I), and Si, Pb and Hg. Alkenes and unsaturated organic compounds can poison palladium as can elemental carbon deposited from decomposition of carbonaceous materials. Carbon monoxide in high concentrations and at low temperatures can form a monolayer which blocks adsorption and dissociation of molecular hydrogen. Carbon monoxide can also decompose to produce car-

bon and CO_2. The carbon can poison the Pd, if not removed. Particulates, including dust or reactor wall corrosion products can encrust membranes. Particulates which adhere to membrane surfaces can act as diffusion point sources of various contaminants which spread, coat and poison membrane catalysts.

In the work on palladium-based membranes of McBride et al. [70], McBride and McKinley [71], and also as quoted by King [72], large guard beds filled with adsorbent were used to remove solid contaminant particles, compounds of sulfur, heavy hydrocarbons, oils, etc. from hydrogen–methane feedstocks. Two guard beds were used in parallel, one remaining in service while the second was reactivated or replaced [70]. A catalytic feedstock conditioner, apparently used to hydrogenate ethylene, propylene, and acetylene impurities in the feed, was also used to avoid poisoning of palladium–silver membranes by unsaturated hydrocarbons [70].

In the mid-1960s, in the days just before escalation of world palladium prices, this system of Union Carbide Corporation (now Praxair) employed 1150 square feet (107 m^2) of Pd–5 wt% Ag membrane, operating between 653 and 663 K (380 and 390 °C), to initially purify 3 000 000 standard cubic feet per day (1 $m^3\ s^{-1}$ STP) of hydrogen to a purity of 99.98% [70]. This was increased to 4 200 000 scfd (1.4 $m^3\ s^{-1}$ STP) using a hydrogen–methane feedstock input pressure between 240 and 580 psig (16.6–40.0 barg) and permeate pressure between 65 and 100 psi (4.5 to 6.9 bar) [71, 72]. Large compressors were used to remove the pure hydrogen from the permeate side of the membranes.

Copper metal compression gaskets were used to provide leak-tight seals in the relatively large, scaled-up system of McBride et al. [70]. The copper gaskets, compressed at room temperature, apparently minimized catalyst poisoning or interdiffusion issues otherwise encountered with high-temperature welding, brazing, or soldering to form membrane seals.

4.9 Catalyst Tolerance to Sulfur

In general, the concentration of sulfur and its compounds would likely need to be reduced to the part per *billion* level to protect unalloyed or palladium–silver-based membranes. Removal of sulfur to tolerable levels from feeds originating from natural gas is much more straightforward relative to removal of sulfur and other impurities from coal according to Riesenfeld and Kohl [73].

McKinley studied the effects of poisoning by H_2S on membrane materials of pure palladium, and alloys of palladium with silver, nickel, copper and gold [74]. Only the Pd–Au alloys showed reasonable resistance to poisoning by hydrogen sulfide according to McKinley [74]. An alloy of 60 wt% Pd and 40 wt% Au retained 80% of its initial (unpoisoned) permeability upon exposure to 4 ppm H_2S and retained 40% of its permeability during exposure to 20 ppm H_2S at 623 K (350 °C). In contrast, pure palladium was poisoned to 30% of its initial, unpoisoned value after 6 days exposure to only 4 ppm H_2S. An alloy of 73 wt% Pd and 27 wt% Ag was completely poisoned within hours after exposure to only 4 ppm

H_2S [74]. An alloy of 60 wt% Pd and 40 wt% Cu fell to about 5% of its initial, unpoisoned permeability after a few days according to McKinley [74].

The Pd–Au and Pd–Cu alloys retained their metallic luster upon exposure to hydrogen sulfide and recovered their activity after removal of H_2S [74], implying that both alloys were resistant to formation of bulk sulfides. The unalloyed Pd and Pd–Ag membranes were much changed in appearance, according to McKinley, indicative of the formation of bulk sulfides. X-ray diffraction analysis at Eltron Research Inc. shows that unalloyed Pd forms the bulk sulfide, Pd_4S, and that Pd–Ag alloys form bulk palladium–silver sulfides, leading to failure of membranes exposed to under 20 ppmv H_2S in the range 593–713 K (320–440 °C) [75]. The Pd 60–Cu 40 (wt%) alloys resist formation of bulk sulfides, but were not completely satisfactory for use with H_2S [75], in accord with the work of McKinley [74].

According to a later patent by McKinley [76], the 60 wt% Pd–40 wt% Cu composition allowed the highest (unpoisoned) hydrogen flux of the palladium–copper alloys tested. The 60–40 Pd–Cu alloy is just slightly removed from the 62.5 wt% Pd–37.5 wt% Cu alloy which gives an atom% of 50–50 Pd–Cu and an atomic ratio of Pd : Cu of 1 : 1. Below 871 K (598 °C) for palladium alloys between approximately 30 and 60 atom% copper, the palladium–copper alloys assume a body centered cubic structure, rather than the usual face centered cubic structure [76–78]. According to Savitsky et al., a mixture of stoichiometric compounds of $PdCu_3$ and Pd_1Cu_1 may exist in this region of the Pd-Cu phase diagram as evidenced by changes in specific heat characteristic for intermetallic compounds [79].

Morreale et al. find that 80 wt% Pd–20 wt% Cu alloys exhibit higher resistance to H_2S (at very high partial pressures of hydrogen) relative to the 60 wt% Pd–40 wt% Cu composition [77]. However, the 80–20 composition has a lower initial (unpoisoned) permeability relative to the 60–40 composition. The 80–20 Pd–Cu composition has the face centered cubic crystal structure as do unalloyed elemental palladium and copper [77].

4.10 Interdiffusion

Interdiffusion between the components of catalyst coatings and substrates can also lead to catalyst deactivation. If Nb, Ta, Ti, V or Zr diffuses through palladium or other noble metal protective layers and reacts on the outer surface to form stable oxides, carbides or nitrides, the catalytic dissociation of molecular hydrogen can be poisoned. Interdiffusion, linked to loss of hydrogen flux, has been reported by Edlund and McCarthy [47] and Paglieri et al. [56]. Membrane surfaces can also become depleted of palladium if palladium diffuses into the substrates. Apparent complete loss of palladium has been observed by Rothenberger et al. for 40 nm thick Pd films on Ta foils after 48 h use at 1173 K (900 °C) [41].

In earlier work, Buxbaum and Marker had used thicker Pd films (1–2 μm) to compensate for the anticipated diffusion of palladium into substrates over the desired lifetime of the membrane [42]. In later work, Buxbaum patented the use

of vanadium–palladium alloys (Pd = 1 to 10 wt%), and niobium–palladium alloys, (Pd = 1 to 10 wt%) [44, 45], among many others, in order to partially saturate the substrates with palladium, and thus eliminate the driving force for diffusion of palladium into the substrates. In a patent of Behr et al., the authors stated use of Pd–Cu, Pd–Ag and Pd–Y catalysts with Nb, Ta, Ti and V substrates which were saturated with Cu, Ag or Y to reduce the driving force for interdiffusion [13]. Tantalum foils, saturated with copper, were coated with 1 μm thick films of 60 wt% Pd–40 wt% Cu and remained intact after 4000 h (167 days) at 673 K (400 °C) according to the inventors [13]. The authors claim that it was impossible to find any intermetallic diffusion between Cu–Pd and their patented alloy substrates at 873 K (600 °C) [13]. In general, the use of alloys or intermetallic compounds in both the catalyst layers and substrates will lower the Gibbs' energy of the system and thus the driving force for metallic interdiffusion [80].

It should be noted that hydrogen plays a critical role in the interdiffusion of palladium-based catalysts and their refractory Group IVB–VB substrates. In the absence of hydrogen, interdiffusion may be undetectable, even at relatively high temperatures. In the presence of hydrogen, interdiffusion may occur at temperatures hundreds of degrees lower than might be predicted for these refractory materials.

4.11
Measured Hydrogen Permeability of Bulk Membrane Materials

For hydrogen partial pressures in feeds of above a few torr, and for thicker membranes (> 250 μm) of Nb, Ta, Ti, V and Zr, hydrogen flux can become limited by bulk diffusion, and, in practice, superpermeability is not achieved. Nevertheless, these elements do possess the highest known permeabilities for hydrogen of any of the common elements, far superior to those of palladium and its alloys, the standards in the hydrogen purification membrane industry.

When diffusion of dissociated hydrogen through dense bulk membrane material is the rate limiting step, hydrogen flux follows Sieverts' Law: $J = (P_e/l)(P_f^{1/2} - P_s^{1/2})$, where J is the hydrogen flux in mol m^{-2} s^{-1}, P_e is the permeability at a specific temperature in mol m m^{-2} s^{-1} $Pa^{-0.5}$ or mol m^{-1} s^{-1} $Pa^{-0.5}$, l is the membrane thickness in m, and P_f and P_s are the partial pressures in Pa of hydrogen on the feed and sink sides of the membrane, respectively. The permeability, P_e, of a material at a specific temperature can be determined if the hydrogen flux, J, hydrogen partial pressures, P_f and P_s, and membrane thickness, l, are measured. Once the permeability at a given temperature is known, the hydrogen flux can be predicted as a function of the membrane thickness and the partial pressures of hydrogen on the two sides of the membrane.

To calculate the hydrogen flux as a function of temperature, permeabilities at various temperatures are determined to derive an Arrhenius expression of the form: $P_e = P_0 e^{-Eact/RT}$ [81, 82]. The pre-exponential factor, P_0, and the activation energy, E_{act}, are derived from the y-intercept and slope of Arrhenius plots of $\ln[P_e]$

vs. $1/T$ [81, 82], respectively. The permeability is equal to the product of hydrogen diffusivity, D (in $m^2\ s^{-1}$) and hydrogen solubility, S (in mol $m^{-3}\ Pa^{-0.5}$). Because the solubility of hydrogen in Nb, Ta and V increases at lower temperature, more rapidly than D decreases, permeability of these elements increases at lower T, as shown in Fig. 4.1, and the apparent activation energies are negative.

As shown in Fig. 4.1, hydrogen permeabilities of Nb, Ta and V far exceed that of Pd. Although Steward has incorporated values of hydrogen diffusivity and solubility from diverse literature sources [4], his derived equations for hydrogen permeability still appear to be one of the best guides available for predicting achievable goals for hydrogen flux using these membrane materials.

Figure 4.1 does not plot values for zirconium or the body centered cubic phase of titanium, because these were not included in the original work of Steward [4]. Permeability values for zirconium, which can exceed those of niobium in much of the temperature range shown, are plotted in the publications of Buxbaum et al. [15, 35, 42] and also those of the group at Los Alamos [53, 54]. Hill plots permeability values for his preferred body centered cubic alloy of titanium (Ti–13V–11Cr–3Al), showing its superior performance to unalloyed Pd in the range 573–723 K (300–450 °C) [12]. The plots of Steward's equations shown in Fig. 4.1, differ from some of those previously published [15, 35, 42, 53, 54] in not extrapolating beyond the limits of Steward's equations and in choosing additional elements of catalytic interest, while eliminating a few materials which are of less interest for hydrogen permeable membranes.

Calculated values of Steward have been verified or approached in a number of cases, as is summarized in Table 4.2. However, it should be stressed that the values predicted by Steward are, in general, not achieved if the flux is limited by gas

Tab. 4.2 Measured values of hydrogen permeability for membranes based on Group IVB–VB elements and their alloys.

Membrane material	Temperature (°C)	Permeability (mol $m^{-1}\ s^{-1}\ Pa^{-0.5}$)	Ref.
Pd coated Ta	350–420	$1.07–1.45 \times 10^{-7}$	41
Pd coated Nb	425	3.2×10^{-7}	42
Pd coated Zr	425	2.2×10^{-6}	42
Pd/Group IV–VB Composite	440	3.2×10^{-7}	81
Pd coated V	300	1.48×10^{-7} [a]	54
Pd coated V	250	5.0×10^{-7}	59
Pd coated V–10Al	250	2.0×10^{-7}	64
Pd coated V–10.5Ni–4.5Al	350	6.29×10^{-8}	63

a Calculated from reported data.

phase diffusion, surface catalysis, diffusion through the catalytic layers, or diffusion through impurities at the catalyst/substrate interface. Necessary precautions must be taken in fabricating, operating, and protecting the composite membranes from impurities.

Buxbaum and the group at NETL have reported permeabilities for palladium coated tantalum of between 1.07×10^{-7} and 1.45×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ for temperatures 623–693 K (350–420 °C) [41]. Weekly backflushes with hydrogen to remove coke and other impurities were recommended [35]. Buxbaum and Marker have reported a permeability for Pd coated Nb of 3.2×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ at 698 K (425 °C), which was only one-third of the theoretical value due to gas phase resistance [42]. The Buxbaum group has achieved a permeability for Zr of 2.2×10^{-6} mol m^{-1} s^{-1} $Pa^{-0.5}$ at 698 K (425 °C) [42]. They state that hydrogen transport through zirconium is 100 times better than through Pd under similar conditions [42].

Moss et al. achieved a permeability for palladium coated vanadium of 1.48×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ at 573 K (300 °C) (Mundschau calculated from Moss et al. data) [54]. An operating temperature of 573 K (300 °C) was found optimum and was used to avoid interdiffusion [54]. No interdiffusion was detected by Rutherford backscattering for Pd/V at 573 K (300 °C) after 100 h, according to Moss et al. [54].

The group of Nishimura has reported values as high as 5×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ for unalloyed vanadium at temperatures as low as 523 K (250 °C) [59]. Their vanadium–10 atom% Al alloys, have reached 2.0×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ at 250 °C [64] and their V–10.5Ni–4.5 Al alloys have achieved a permeability of 6.29×10^{-8} mol m^{-1} s^{-1} $Pa^{-0.5}$ at 623 K (350 °C) [63].

Although the vanadium alloys of Nishimura et al., have lower hydrogen permeability than pure vanadium, the alloys reduce the susceptibility of vanadium to both hydrogen and oxygen embrittlement [59]. As with Pd–Ag alloys, pre-expansion of the crystal lattice of vanadium by alloying with other elements [83] may possibly improve the mechanical properties of the membranes by reducing effects of chemical expansion by absorption of hydrogen. However, expansion of the vanadium lattice also increases the hopping distance between interstitial sites, which, in general, will reduce hydrogen diffusivity [84]. The permeability of the alloy, V–10.5Ni–4.5 Al is twice that of V–15Ni [64]. Baking in air recovered losses in permeability over long-term tests [64].

The lattice of vanadium expands approximately linearly with the addition of aluminum [64]. The aluminum intermetallic compound, V_3Al (V–25 atom% Al), expands the lattice by about 1% from 0.3025 nm in unalloyed vanadium to 0.3054 nm [64]. Molybdenum, cobalt and titanium also expand the lattice of vanadium, whereas elements such as chromium and iron cause the lattice to contract [83]. Addition of these elements can increase the mechanical strength of alloys relative to unalloyed vanadium [85]. For niobium and tantalum, mechanical properties can also be improved by alloying [86]. Buxbaum has patented a number of alloys of niobium, tantalum and vanadium for membrane use, including Ta–W, V–Co, V–Pd, V–Au, V–Cu, V–Al, Nb–Ag, Nb–Pt, Nb–Pd, V–Ni–Co, V–Ni–Pd, V–Nb–Pt, and V–Pd–Au [45].

Mundschau et al. have achieved values of 3.2×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ for their Groups IVB–VB based composite materials [81]. Figure 4.3 shows Sieverts' Law plots of hydrogen flux vs. the difference in the square roots of the hydrogen partial pressures on membrane feed and sink sides, for Group IVB–VB composite membranes tested at Eltron Research Inc. To avoid complications of possible poisoning of catalysts, the membranes were initially examined under ideal hydrogen–helium gas mixtures with hydrogen partial pressure up to 13.1 bar and total pressure up to 34 bar. The membranes were supported only at the rim by metal gaskets, similar to that described by Peachey et al. [53], except that stainless steel was replaced by INCONEL 600 alloys, which were better suited for studies with high pressure steam and other components of water-gas shift mixtures. Helium was used to check for leaks. No leaks to helium were detected, which is typical for such sealing methods. Membrane thicknesses of 500, 250, 127 and 75 microns were studied at

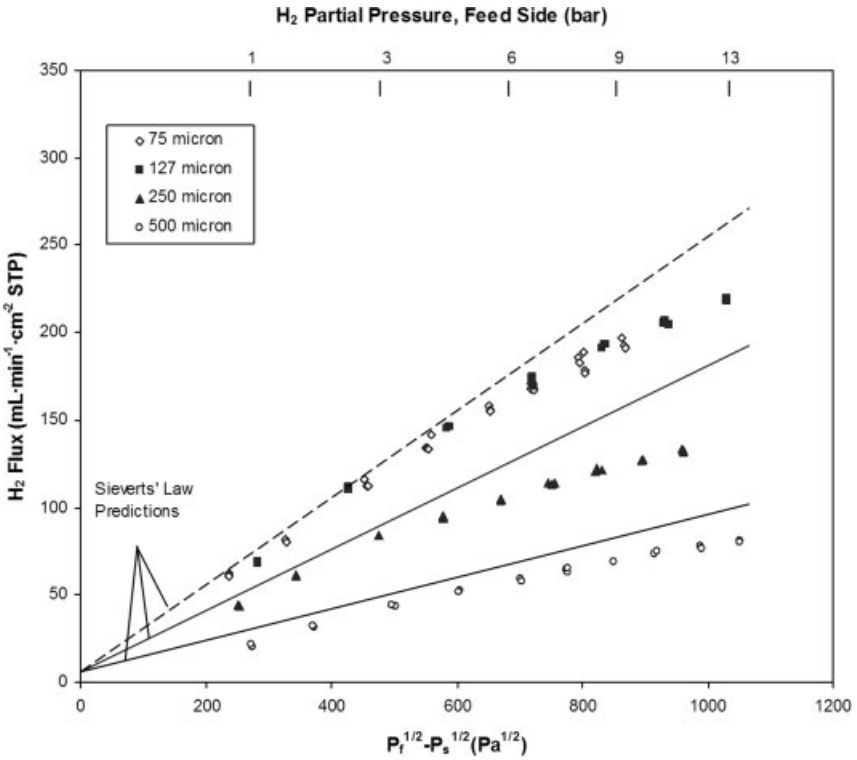


Figure 4.3 Hydrogen flux through membranes of various thickness fabricated with Group IVB–VB elements. Data show that the rate limiting steps determining hydrogen flux vary and include diffusion through the bulk membrane material for the thicker membranes, gas phase diffusion at high pressures, and the effects of the catalytic layers for the thinnest membranes. Data was acquired at 713 K (440 °C), under ideal 40 mol% hydrogen/60 mol% helium feed conditions with 100% selectivity towards hydrogen and a differential pressure up to 33 bar.

713 K (440 °C), which is the upper temperature tolerated by commercial Fe_3O_4/Cr_2O_3 high-temperature water-gas shift catalysts employed in the study [87]. Research goals were to test membranes under simulated high-temperature water-gas shift reactor conditions, 613–713 K (340–440 °C), and to resist differential pressures over 30 bar [81]. Palladium-based catalyst layers were approximately 400 nm on each side.

For the 500 μm thick membrane, the thickest membrane employed, data falls fairly well on the Sieverts' Law line, which is interpreted as implying that bulk diffusion of dissociated hydrogen is the rate limiting step. Experimentally measured permeability for the 500 μm membrane was 3.2×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$. As the higher feed pressures were approached (13.1 bar partial pressure hydrogen, 34 bar total pressure with helium), some deviations from Sieverts' Law were observed. These are attributed to gas phase diffusion limitations, as discussed below.

When the membrane thickness was halved, from 500 μm to 250 μm, the hydrogen flux doubled, in accord with Sieverts' Law, and the measured permeability remained at 3.2×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$, implying that bulk diffusion was the rate limiting step at lower feed pressures and that this value of permeability was characteristic of the bulk membrane material. At higher feed pressures, larger deviations from Sieverts' Law were seen, which is again attributed to gas phase diffusion limitations. Evidence for this is shown in Fig. 4.4. If the concentration of hydrogen in the feed is increased from 40 mol% to 60, 80 or 100 mol%, the data

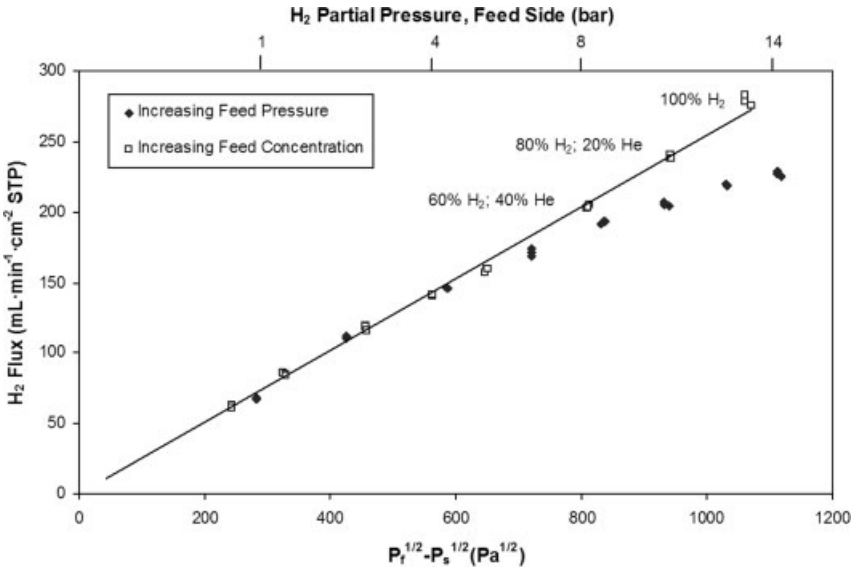


Figure 4.4 Data showing that observed deviations from Sieverts' Law in membranes with very high permeability arise from limitations due to gas phase diffusion in the hydrogen feed. By increasing concentrations of hydrogen in the feed, the data returns to the Sieverts' Law line.

falls back onto the Sieverts' Law line. When the gas delivery and turbulence were improved, the data also fell back onto the Sieverts' Law line in accord with the views of Buxbaum [32]. This behavior is consistent with the likely presence of a stagnant layer of hydrogen depleted gas above the membrane surface.

In the limit for pure gas phase diffusion, which follows Fick's First Law of Diffusion, the gas phase hydrogen flux, J, is given by $J = dc/dx$, where dc is the change in concentration over the distance dx [14]. According to Fick's First Law of Diffusion, the hydrogen flux will increase linearly with hydrogen concentration and will be directly proportional to the difference in hydrogen partial pressure. If diffusion through the membrane material is the rate limiting step, Sieverts' Law should be followed, and the pressure dependence will be proportional to the square root of the hydrogen partial pressure, $p^{0.5}$. In the regime where gas phase diffusion and diffusion through the dense bulk material show comparable transport resistance, the pressure exponent, p^n, should lie between $p^{0.5}$, for pure Sieverts' Law behavior, and p^1, for pure gas phase diffusion.

Additional evidence for gas phase diffusion limitation is very small or negligible apparent activation energy, which can be derived from an Arrhenius plot of $\ln[P_e]$ vs. $1/T$. For a gas phase diffusion limitation, increase in hydrogen flux should be negligible over small ranges of increase in temperature. In the water-gas shift temperature range of interest, approximately 613–713 K (340–440 °C), the velocity of molecular hydrogen should increase by a factor of $(713\ K/613\ K)^{1/2} = 1.08$, a relatively negligible amount. Simultaneous slight decrease in gas density over this temperature range will occur, decreasing the incident flux of hydrogen from the gas phase onto the membrane surface.

When bulk diffusion through the palladium catalyst layers is rate limiting, an activation energy near 15.6 kJ mol^{-1} is expected according to Steward [4]. Work of Mundschau et al. [81] found an activation energy for a 100 μm thick unalloyed palladium foil to be 15.7 kJ mol^{-1}, in good agreement with the theoretical work of Steward. Basile et al. summarize some palladium activation energies from the literature [82]. For a 200 μm unalloyed palladium they list an activation energy of 18.45 kJ mol^{-1}, consistent with the above values. Wipf gives a value of 0.23 eV = 22.189 kJ mol^{-1} for palladium [84]. However, for quoted work of Peachey and Dye for 1 μm of unalloyed Pd on tantalum, they list an activation energy of only 5.3 kJ mol^{-1}, which may imply a rate limiting step other than diffusion through bulk palladium [82].

If bulk diffusion through Nb, Ta or V is rate limiting, activation energies can be negative (i.e. hydrogen flux increases as temperature decreases due to significantly increased solubility of hydrogen in the metals). Negative activation energy for Nb, Ta and V follows from Fig. 4.1 and the work of Steward – if bulk diffusion is the rate limiting step. A negative activation energy of –17.7 kJ mol^{-1} for Group IVB–VB membrane material in the temperature range 593–653 K (320–380 °C) has been reported by Mundschau et al. [81], in accord with the work of Steward [4].

For further decrease in membrane thickness from 250 to 127 μm, the flux data shown in Fig. 4.3 no longer increase as predicted by Sieverts' Law, and the permeability decreases to 2.4×10^{-7} mol m^{-1} s^{-1} Pa$^{-0.5}$. Arrhenius behavior also changes

from limitations due to bulk phase diffusion through Group IVB–VB based materials with an activation energy of –17.7 kJ mol^{-1}, to an activation energy of +11.5 kJ mol^{-1}, which is more consistent with limitations due to bulk diffusion through the palladium-based catalytic overlayers [81].

Finally, when the membrane thickness is reduced from 127 to 75 μm, as shown in Fig. 4.3, the permeability drops to 1.3×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ and no increase in hydrogen flux is observed. The lack of increase in the hydrogen flux as the membrane thickness is reduced, demonstrates unambiguously that diffusion through the bulk membrane material is no longer rate limiting at a thickness of 75 μm, but that bulk diffusion through the palladium-based catalysts, or other effect involving the catalytic layers, becomes rate limiting at lower feed pressures. At higher feed pressures, deviations from Sieverts' Law again show limitations due to gas phase diffusion. The data of Fig. 4.3 indicate that for thinner membranes (<250 μm) of Group IVB–VB materials, the hydrogen flux can easily become limited by gas phase diffusion as well as effects involving the catalytic layers.

Figure 4.5 shows hydrogen flux data of Eltron Research Inc., again for an ideal hydrogen–helium feed mixture, for a membrane of Group IVB–VB material

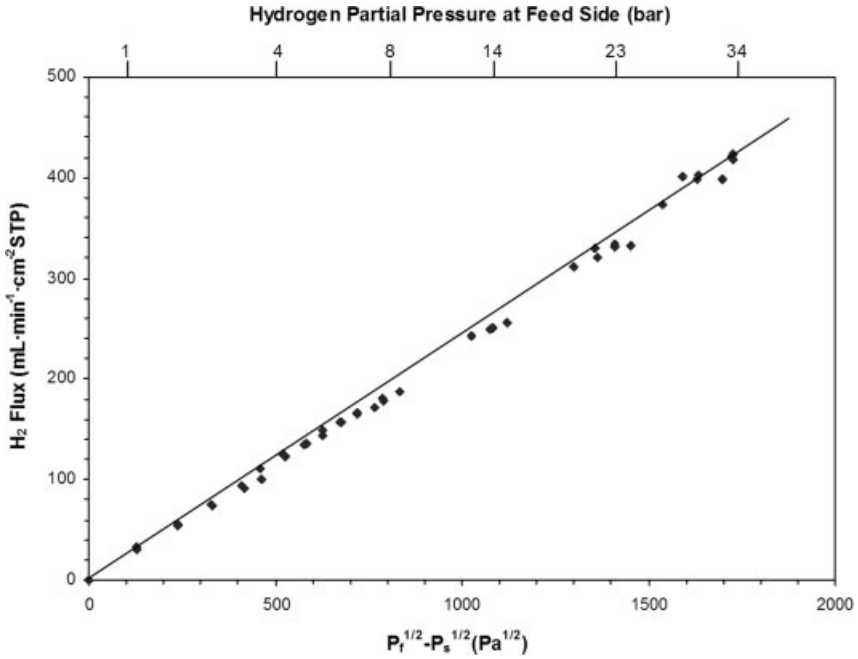


Figure 4.5 Experimental data showing hydrogen flux of over 420 (mL H_2) min^{-1} cm^{-2} (STP) (3.15 (mol H_2) s^{-1} m^{-2}) at 713 K (440 °C) using an ideal H_2/He feed mixture with up to 34 bar partial pressure of H_2 and slightly over 33 bar differential pressure with approximately 0.1 bar partial pressure of H_2 in the argon sweep. Permeability was 2.3×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$. Selectivity was 100% for hydrogen, with no detected leak of helium.

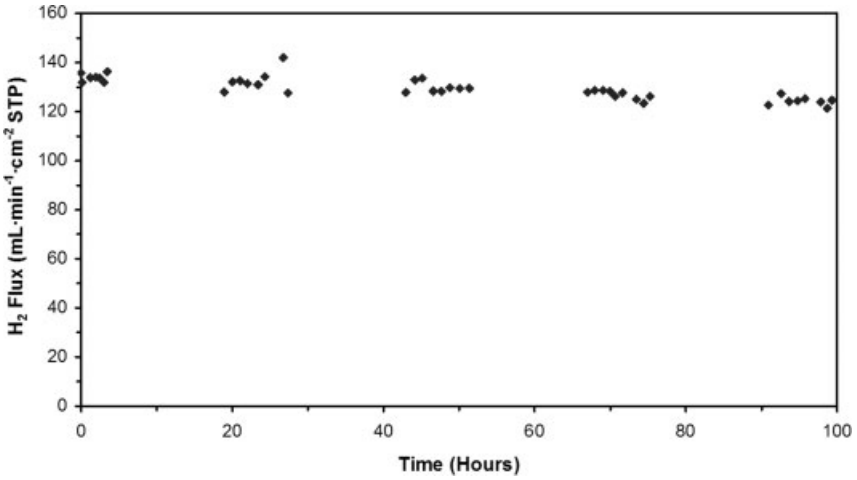


Figure 4.6 Composite membrane tested at 673 K (400 °C) with an ideal hydrogen/helium feed, a partial pressure of hydrogen of 8.3 bar (120 psi) and total feed pressure of 20.7 bar (300 psi). No sweep gas was used, and the permeate pressure of pure hydrogen was maintained at 0.85 bar.

127 μm (0.005 inch) thick, also coated on both sides, each with approximately 400 nm of palladium-based catalyst. Gas phase diffusion limitations at high pressure were overcome by better gas mixing and use of higher concentrations of hydrogen in the gas mix, reaching almost pure hydrogen at 34 bar total feed pressure. With elimination of gas phase diffusion limitations, Sieverts' Law was followed very well, and the square root dependence on partial pressure of hydrogen was seen. This should be the case if molecular hydrogen is dissociated and if bulk diffusion through the dense metal is rate limiting. Permeability of this membrane at 713 K (440 °C) was 2.3×10^{-7} mol m^{-1} s^{-1} Pa$^{-0.5}$, which was in the regime where diffusion was partially limited by the relatively thick catalytic layers (400 nm on each side or 800 nm total). Argon was used as a sweep gas to maintain total pressure near 1 bar and partial pressure of hydrogen in the permeate at approximately 0.1 bar. The membrane was successfully tested to a differential pressure of 33 bar.

Figure 4.6 shows hydrogen flux data for a membrane of Group IVB–VB material tested at 673 K (400 °C), with no sweep gas. Hydrogen partial pressure in the feed was 120 psi (8.3 bar). Total pressure in the feed was 300 psi (20 bar), the balance being helium. Hydrogen pressure in the pure hydrogen permeate was at approximately 0.85 bar. The permeability of this 127 μm thick membrane was 2.16×10^{-7} mol m^{-1} s^{-1} Pa$^{-0.5}$. Permeability was partially limited by the catalytic layers.

Figure 4.7 shows hydrogen flux data for membranes of Group IVB–VB material tested at Eltron Research Inc. using feeds containing various components of a water-gas shift mixture. The membranes were tested with a total pressure in the feed up to 450 psi (31.0 bar) and with argon sweep gas at ambient pressure. In an

ideal mixture containing 40 mol% H_2 and 60 mol% He, Sieverts' Law is followed at lower pressures (<200 psig or 13.8 barg), but limitations due to gas phase diffusion are reached, and ultimately limit membrane performance. Upon replacement of most of the helium with steam, flux behavior is basically identical to that of the ideal hydrogen–helium mix, showing that steam does not adversely affect the palladium-based catalysts. It should be pointed out that special care was taken to ensure that all components of the reactor exposed to steam were fairly inert to corrosion. If this precaution is not taken, the components of the reactor walls can be hydrothermally transported to the membrane surfaces, poisoning the catalysts. In addition, guard beds of high surface area adsorbent were used to protect the membrane catalysts from poisons and particulates.

Using a mixture containing 37% H_2O, 18% CO_2 and 41% H_2 by volume with a balance of helium, little change relative to the ideal conditions was noted. Poisoning of palladium by CO_2 is not expected because palladium does not form carbonates, and CO_2 readily desorbs from palladium at 693 K (420 °C). It should be noted that mixtures of steam and CO_2 can hydrothermally transport many materials, such as silicon, which can poison membrane catalysts if improper reactor wall

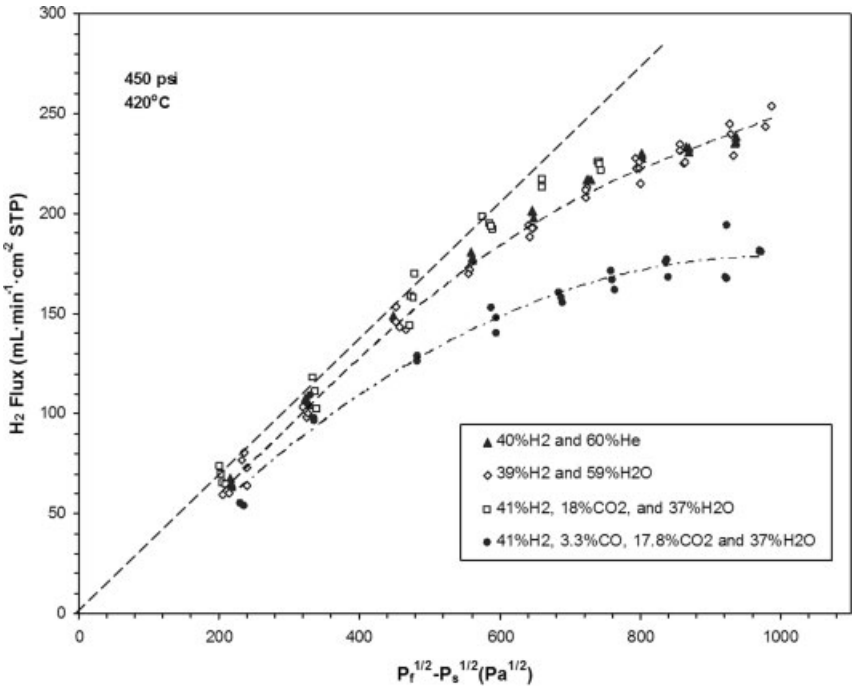


Figure 4.7 Hydrogen flux upon exposure to various components of a high-temperature water-gas shift mixture. Steam and CO_2 do not affect flux or catalysts relative to an ideal H_2/He mixture. Some loss in hydrogen flux occurs with addition of CO, but a flux of 150 mL min^{-1} cm^{-2} (STP) was achieved with over 1 bar CO in the mix at 693 K (420 °C).

components are chosen or if the membranes are not protected by guard bed adsorbents.

Figure 4.7 shows that the presence of 3.3 mol% CO partially inhibits hydrogen flux, maximum values dropping from about 250 mL min^{-1} cm^{-2} (STP) to 150 mL min^{-1} cm^{-2} (STP) with 1 bar CO added to the system. This is attributed to competitive adsorption of CO with H_2 for surface sites [75]. Hydrogen flux, however, remains quite respectable despite 1 bar CO in the mixtures. This is in accord with the work of Buxbaum et al., who found no signs of catalyst poisoning or membrane hydrogen embrittlement in a synthesis gas mixture of H_2 + CO for palladium coated V–4Cr–4Ti and V–7.5Cr–15Ti membranes [88]. At much lower temperatures, adsorption of CO can form a complete monolayer on palladium catalysts, completely poisoning hydrogen dissociation [75].

Figure 4.8 shows the results of a long-term study at Eltron Research Inc. in a water-gas shift mixture containing 41.4 mol% H_2, 3.3 mol% CO, 17.8 mol% CO_2, and 37.3 mol% steam with a balance of inert gases. The study was conducted at ambient pressure for over 2800 h (4 months) under continuous operation in the water-gas shift mixture. Membrane catalysts were protected by guard bed adsorbents. After four months, a permeability of 1.0×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ was maintained.

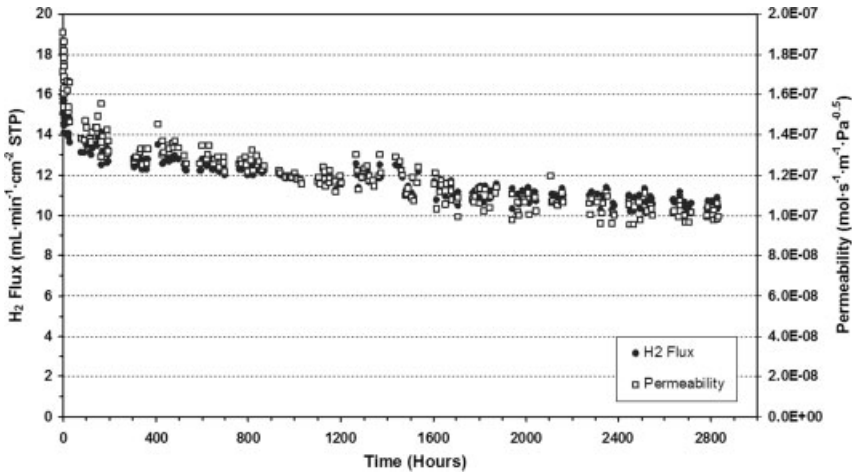


Figure 4.8 Long-term test in a water-gas shift mixture at 613 K (340 °C) and ambient pressure. A permeability of over 1×10^{-7} mol m^{-1} s^{-1} $Pa^{-0.5}$ remained after over 2800 h (4 months) continuous operation in the mixture. Guard bed adsorbents were used to protect the membrane catalysts.

4.12 Conclusions

Superpermeability for hydrogen, in which every hydrogen atom incident upon a membrane surface is transported through the membrane, has been achieved only under very special conditions, such as in the extraction of hydrogen from plasmas, in which the initial flux of hydrogen incident on the membranes is relatively low. Nevertheless, composite membranes based upon niobium, tantalum and vanadium have demonstrated exceptionally high permeabilities with 100% selectivity for hydrogen, far exceeding those of palladium and its alloys. New alloys and methods of fabrication and scale-up may make the membranes practical in industrial applications [89].

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